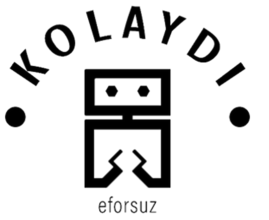


Autonomous Unmanned Ground Vehicle

Shareholders: Mehmet Taha Bekar, Enfal Çabuk, Yunus Talha Erzurumlu, Burcu Şakır, Duru Uyar
Advisor: Assoc. Prof. Dr. Eren Balevi



Project Description

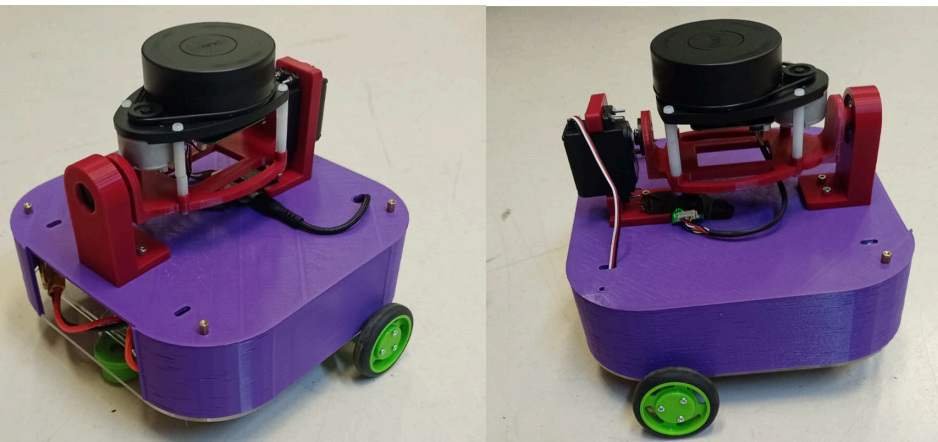
- Autonomous UGV, designed and manufactured by **Kolaydi**, is a robot that can produce a **3D map** of an indoor environment using **2D LIDAR scans**.
- The robot can **localize** itself on the map it generates.
- The robot can go to any place **autonomously** in the room. Its path-planning capabilities are not restricted to line-of-sight places.
- It can adjust its path when there is a **dynamic object** or object not present in the map.
- The robot can be controlled by a computer, and the commands for moving to a place are given by the computer. The map, the planned path, and the robot's location can be observed on the computer. A **user interface** is established for all of the control parts of the computer.
- The robot and the computer communicate via wireless communication. All data transmission is achieved **wirelessly**.

Technical Specification

- The robot's height is ≤ 30 cm.
- The wireless communication between the robot and the user's computer must be established.
- There must be a user interface to control the robot.
- LIDAR rotation precision should be under 2 degrees.
- Map pieces obtained from different grids must be merged.
- Mapping error $\leq 10\%$.
- Localization error ≤ 5 cm.
- Being able to plan a path and **navigate** through that path **autonomously**.
- Being able to reach a given target point with an error ≤ 20 cm.
- Stationary obstacles** should be included in path planning.
- The path needs to be updated when a **dynamic obstacle** is detected.
- Operation time in an environment of a maximum of 8 m² with moderate obstacles ≤ 40 minutes.

Deliverables

- The Robot:**
20 cm diameter, 18 cm height, 1 kg weight
- User Interface Software:** Rviz
- User Guide**
- Test Results**

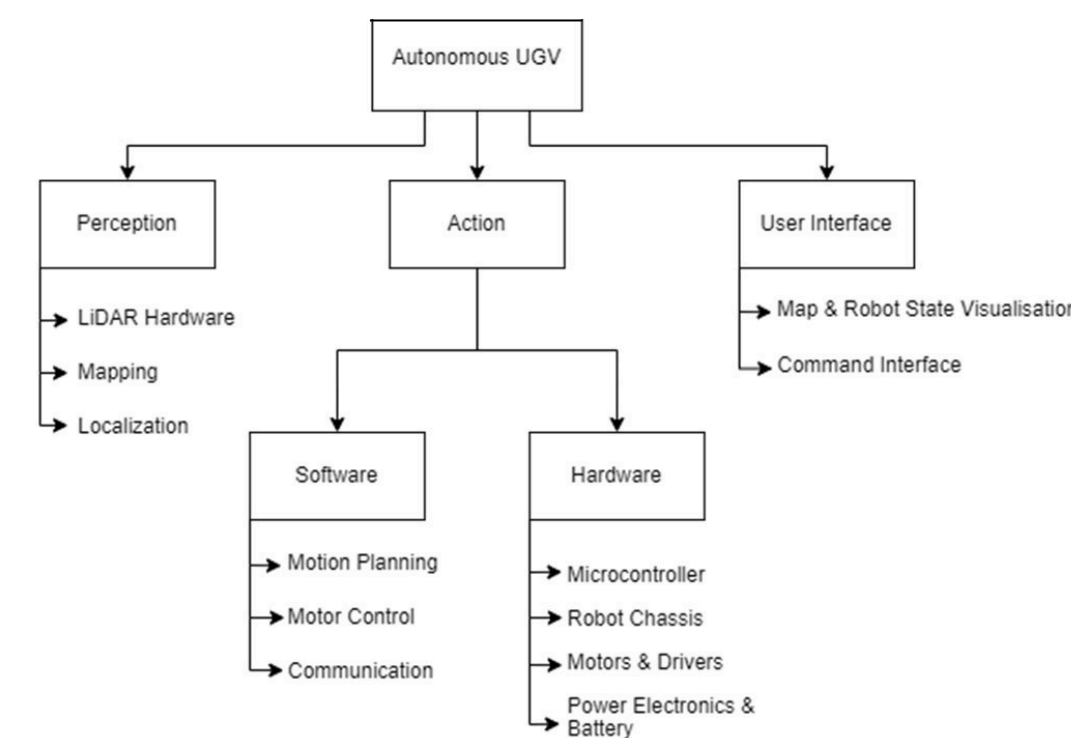


Final Product

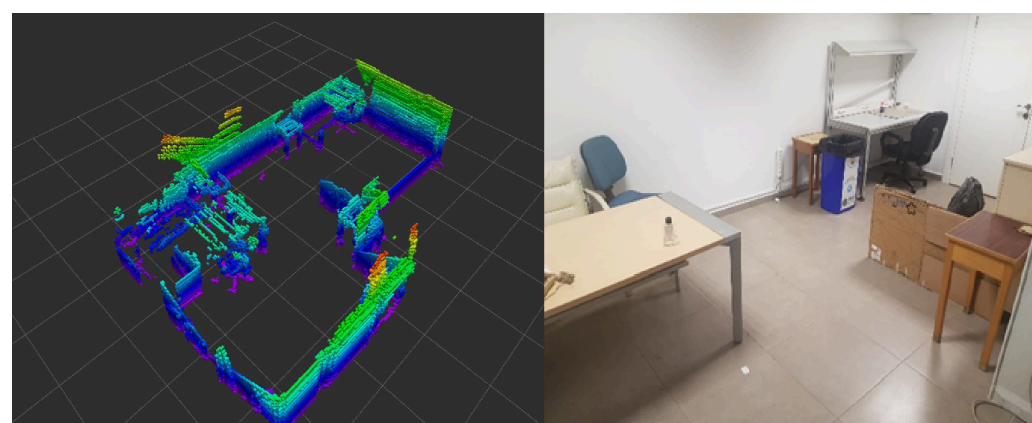
Solution

Our solution includes 3 main parts: **Perception**, **Action**, and **User Interface**.

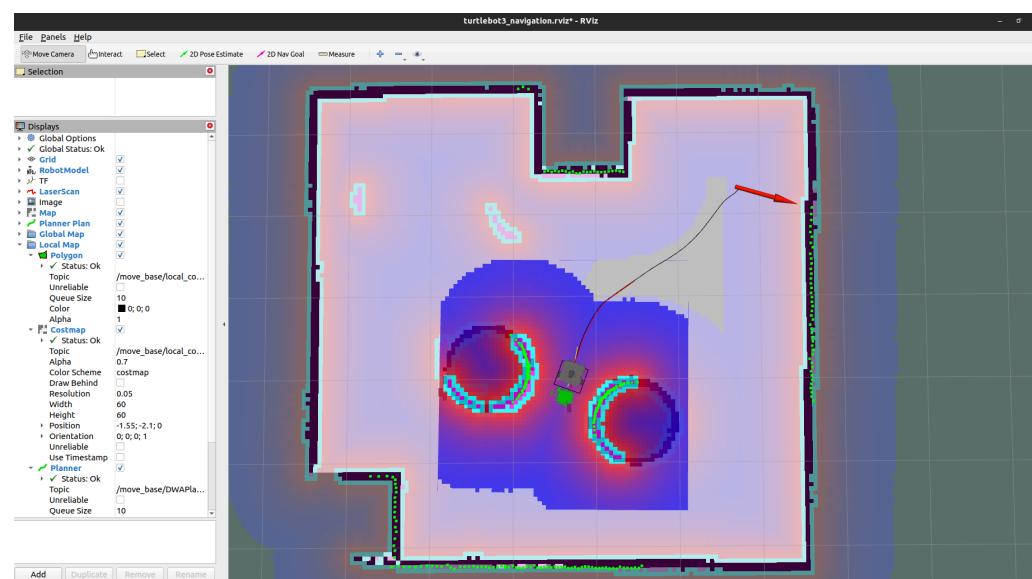
- The perception part includes all the hardware related to the LIDAR, the **mapping** algorithm, and the **localization** algorithm.
- The action part consists of two subparts: software and hardware. The software includes **motion planning**, **motion control**, and **communication** between the robot and the user computer. The hardware includes the microcontroller, the chassis, the motors and drivers, and the battery.
- The user interface includes a visual of the map, the robot's state, and a command interface.



Solution Block Diagram



Mapping

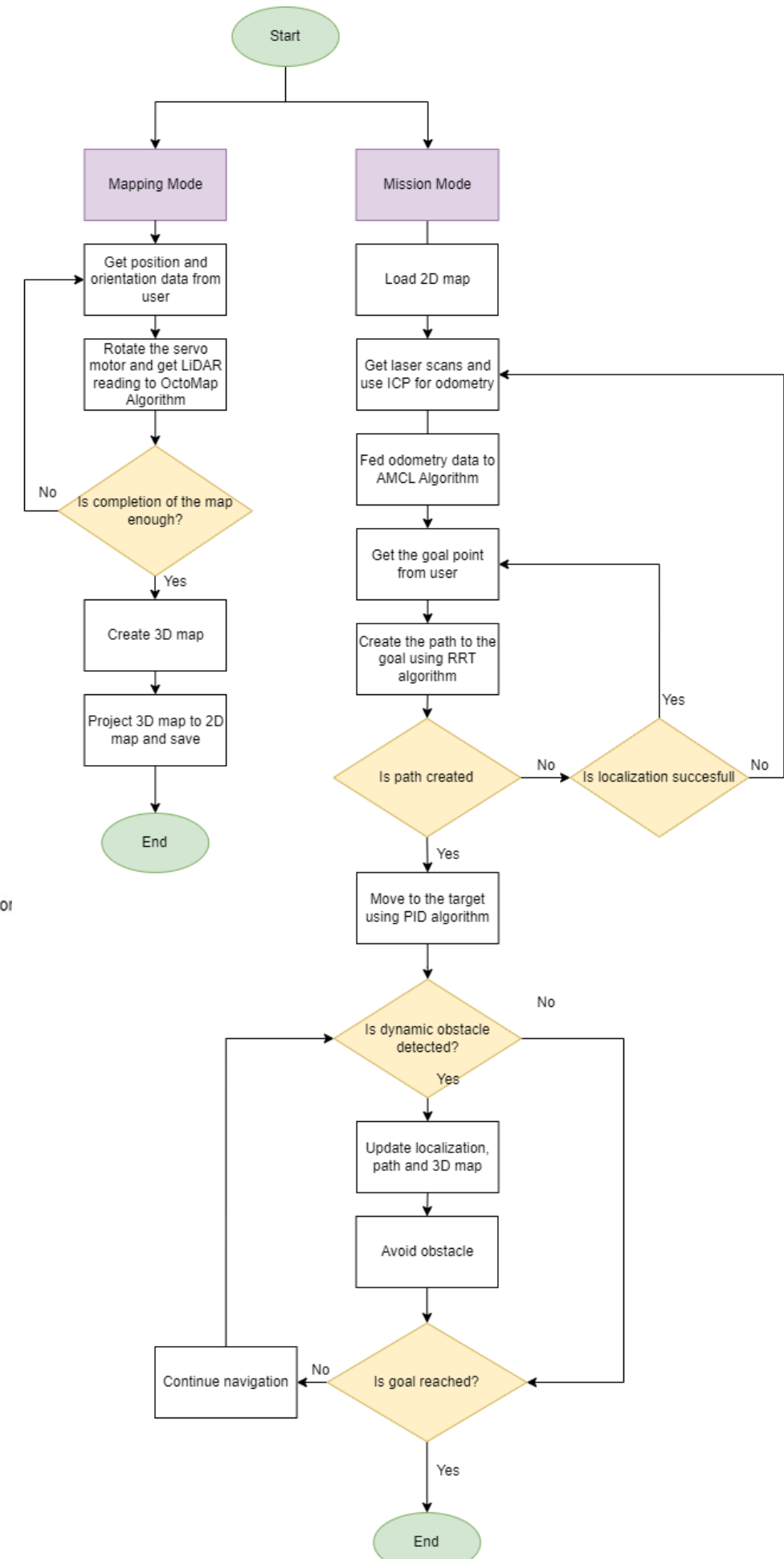


User Interface

Cost Analysis

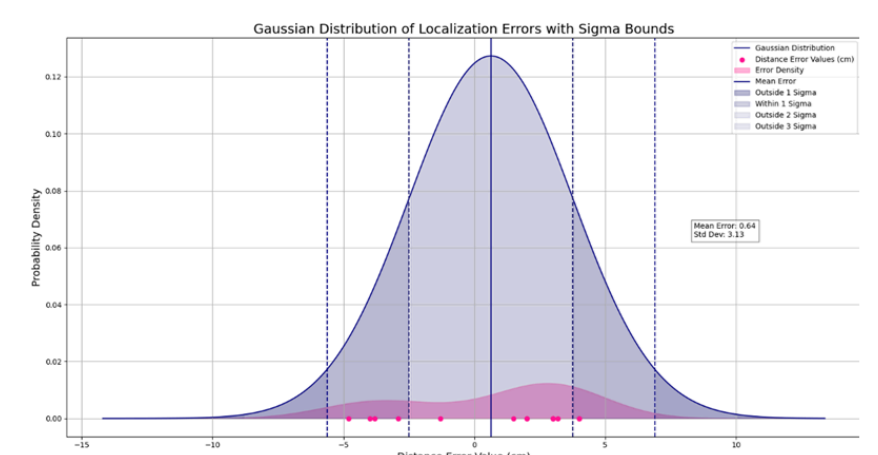
Product	Cost
RPLiDAR A1M8	100\$
Raspberry Pi 4 4GB	69.09 \$
TB6612FNG Motor Controller	2.80 \$
Battery	30.69\$
Robot Chassis with Wheels	9.88\$
2 x DC Motor	3.91\$
Buck Converter	2.67\$
FEETECH FS5103B-FB Servo Motor	13\$
LiDAR Holder	5\$
Total	237.04\$

Design

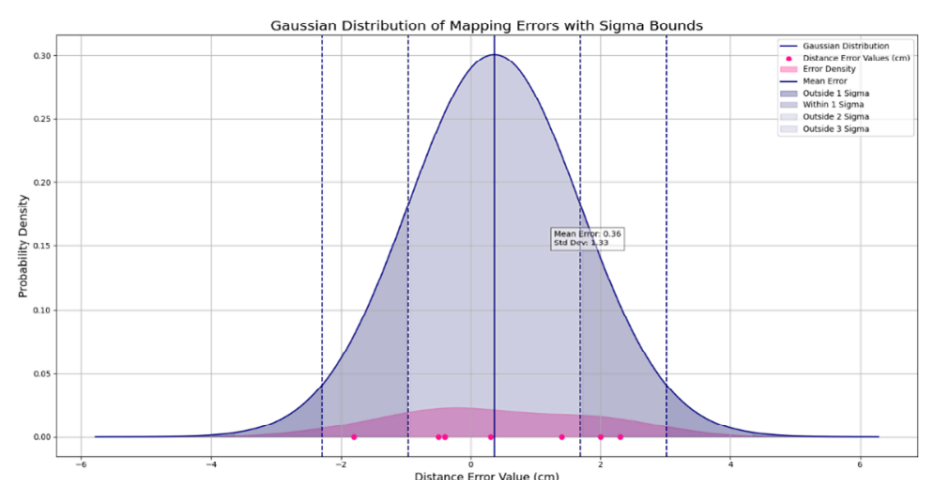


Algorithm Flowchart

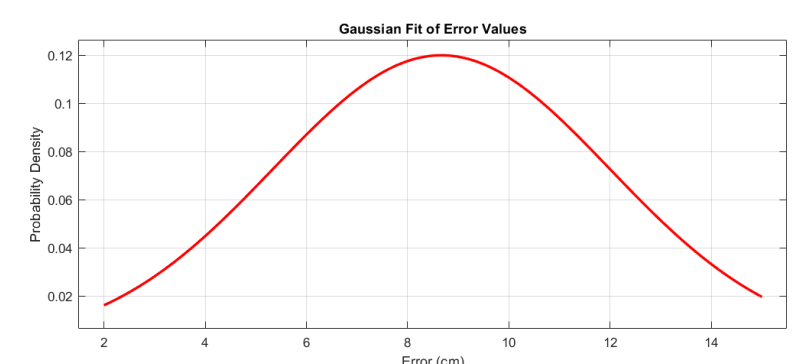
Test Results



Localization Test Results



Mapping Test Results



Motion Planning Test Results